Himanshu Singh

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LinkedIn | Portfolio

Professional Summary

AI SME / Team Lead – AI Engineer with **5 years of experience** delivering **AI and GenAI solutions** for **U.S. healthcare and Fortune 100 clients**. Experienced in **AI solution architecture, client engagement, AI agents, RAG, OCR, document intelligence, LLM applications**, and workflow automation using **Azure AI Foundry, LangChain, LangGraph, Python, Django, FastAPI, and Azure services**. Proven expertise in leading teams, translating business requirements into scalable AI solutions, and leveraging a strong **data engineering background** across Azure, Databricks, Spark, Hadoop, and large-scale data platforms.

Experience

TATA CONSULTANCY SERVICES TCS Healthcare Innovation Lab (Nov 2025 – Present)
AI SME – TL *Hyderabad, India*

- Leading a team within the **TCS Healthcare Innovation Lab**, partnering with **U.S. healthcare clients** to understand business requirements, design solution architectures, and deliver high-volume **AI solutions** across diverse healthcare use cases.
- Engaging directly with client stakeholders to **gather requirements, translate business problems into technical solutions, and lead end-to-end delivery** in collaboration with the engineering team.
- Architecting and delivering diverse AI applications including **OCR, medical document extraction, intelligent text processing, information extraction, and workflow automation** using **Azure AI Foundry, LangChain**, and Azure AI services.
- Developing scalable full-stack AI applications using **Azure AI Foundry, LangChain, Azure Functions, Django, React**, and Azure cloud services, integrating AI capabilities into enterprise healthcare workflows.
- Delivered a portfolio of **AI-powered healthcare solutions** including **Auto RFP, MR Extractor, handwritten prescription OCR, and discharge bill processing**, along with numerous client-specific AI use cases.
- Driving AI solution development from **client discovery and architecture design through prototyping, development, and delivery**, while guiding the team on AI technologies, solution design, and implementation.

TATA CONSULTANCY SERVICES KAISER PERMANENTE (Mar 2023 – Present)
Data Engineer *Hyderabad, India*

- Delivered **20TB+ of healthcare data** into **Azure Data Lake Storage (ADLS)** by **designing and optimizing enterprise-scale data pipelines** using **Azure Data Factory (ADF)** and **Databricks Delta Lake**, enabling faster analytics and reporting.
- Increased data processing efficiency by **60%** by **automating ETL/ELT workflows** across **ADF, ADLS, Databricks, and PySpark**, ensuring near **real-time data availability**.
- Improved data accuracy by **100%** through **implementing robust validation and quality frameworks**, delivering highly reliable and actionable business insights.
- Enabled seamless enterprise-wide analytics by **integrating on-premise, third-party, and Hadoop data sources** (**Oracle, Taleo, JSON, Avro, CSV, Parquet**) into **ADLS** via **ADF connectors**.
- Reduced Spark job runtime by **20%** for **2M+ daily records** by **implementing Spark performance tuning techniques** (caching, partitioning, broadcast joins) across **Hadoop environments**.
- Optimized **on-premise Hadoop clusters (HDFS, Hive, Spark)** by **fine-tuning resource allocation and data partitioning strategies**, improving large-scale batch processing performance by **35%** and ensuring smooth integration with **Azure cloud migration pipelines**.

TATA CONSULTANCY SERVICES WALGREENS (Oct 2021 – Feb 2023)
Data Engineer *Noida, India*

- Accelerated large-scale text processing by **90%** (2 hours → 15 minutes) by **developing and optimizing MapReduce programs** in the **Hadoop ecosystem**, significantly improving ETL throughput.
- Enhanced operational efficiency by **50%** by **automating data workflows** using **Python**, **HiveQL**, and **Shell scripting**, reducing manual intervention and improving reliability.

- Improved analytical data trustworthiness by **30%** through **data quality, consistency, and reconciliation checks** using **Hive**, **Impala**, and **Spark**.
- Streamlined ETL orchestration and reduced manual efforts by **40%** by **automating complex data pipelines** with **Python**, **HiveQL**, and **Shell**, achieving a **60% improvement in throughput**.
- Supported cloud migration initiatives by **managing data ingestion and batch processing** in **Hadoop Distributed File System (HDFS)**, laying the foundation for **Azure-based architectures**.
- Delivered enterprise-wide reporting solutions by **collaborating with business teams** and **processing large-scale datasets** across distributed ETL systems.

Open Source Contributions

Liu Embeddings [PyPI] [Docs]
Developed and published an open-source Python library, **liuembeddings**, which combines **text embeddings** and **vector storage** for efficient local NLP retrieval tasks. Built on **Hugging Face Transformers** and **ChromaDB**, it offers a **free, fully local, and privacy-friendly** alternative to cloud-based embedding services ideal for **AI projects** requiring fast and cost-efficient **embedding, storage, and retrieval** workflows.

DbMeta [PyPI] [Git]
Developed and published an open-source, zero-dependency Python engine, **DbMeta**, designed for parsing and analyzing batch job metadata from **Hadoop, Spark, Databricks** by using folder-based job-run structures. Automatically detects **metadata** files, extracts structured metadata, and converts them into **queryable tables** for efficient monitoring and analytics.

create-setup [winget] [Git]
Developed a lightweight **Windows PowerShell utility** to create Python virtual environments outside project directories, keeping dependencies separate from the **shared drive**. Supports **one-command environment creation**, multiple environments, custom requirements files, and automated dependency installation.

Technical Skills

AI & GenAI:	Azure AI Foundry, LangChain, LangGraph, Prompt Engineering, Open-Source Models
Backend & APIs:	Python, Django, Django REST Framework, FastAPI
Cloud & Azure:	Azure Functions, Azure Blob Storage, Azure Queue Storage, ADLS, Azure DevOps
Vector & DataStores:	MongoDB, SQL, PostgreSQL, MySQL, Oracle
Data Architecture:	ETL/ELT, Data Modelling, Data Warehousing, Data Lakehouse, OLAP/OLTP
Data Engineering:	Databricks, Apache Spark (PySpark, RDD, Streaming), Hadoop, Hive, Impala, MapReduce, ADF, Azure Synapse

Certifications

Databricks Fundamentals | 2025	View Certificate
Databricks Certified Data Engineer Associate | 2025	View Certificate
Azure Fundamentals (TCS Digital) | 2025	View Certificate
Generative AI (Udemy) | 2025	View Certificate
Python Programming (TCS Digital) | 2022	View Certificate

Projects

DataLemur SQL to PySpark	View on GitHub
Databricks Ecommerce analytics	View on GitHub
AI Chatbot	View on GitHub
The Analysis Bot	View on GitHub
AI Data Engineer Copilot	View on GitHub
AI Power Assistant	View on GitHub
Autonomous ML Engineering Agent	View on GitHub
AI Teams of Agents	View on GitHub

Education

NIET — Greater Noida, UP
Bachelor of Technology in Computer Science & Engineering — *Aug 2017 – Sep 2021*

ABC Public School — Gorakhpur, UP
Senior Secondary School — *Apr 2015 – Mar 2017*